



Lecture 22- Extended ACL (Access Control List)



Common Port Numbers (Quick Memory Table)

Port Protocol Service

20 TCP FTP Data

21 TCP FTP Control

22 TCP SSH

23 TCP Telnet

25 TCP SMTP

53 TCP/UDP DNS

67 UDP DHCP Server

68 UDP DHCP Client

69 UDP TFTP

80 TCP HTTP

110 TCP POP3

123 UDP NTP

143 TCP IMAP

161 UDP SNMP

443 TCP HTTPS

445 TCP SMB

514 UDP Syslog

3389 TCP RDP

Port Protocol Service

3306 TCP MySQL

8080 TCP Alternate HTTP

5900 TCP/UDP Virtual Network Computing (VNC) Remote Frame Buffer (RFB)

💡 Daily-Life Analogy:

Ports = doors in a building 🏢. Each service has its own door.

Without ports → all traffic would crash at the same door, confusion everywhere ❌

🔒 2 What is an Extended ACL?

Extended ACL is a set of rules applied on routers to **control traffic precisely**:

- Not just by IP, but also by **protocol, port number, source, and destination**.
- Provides **more security** and **flexibility** than standard ACL.

🧱 Extended ACL Features

Feature	Description
Filters on	Source IP, Destination IP, Protocol, Port
Placement	Near the source to save bandwidth
Number Range	100–199 (classic), 2000–2699 (expanded)
Flexibility	Can allow/deny TCP, UDP, ICMP, etc.
Purpose	Precise control over which traffic is allowed or blocked

🎯 Why Use Extended ACL?

- Block specific host from accessing a server
- Deny certain protocol traffic (FTP, Telnet, HTTP)

- Control traffic to a specific port only

💡 Example Real-Life Analogy:

Security guard at the entrance checks:

- Who you are (Source IP)
- Where you are going (Destination IP)
- What you are carrying (Protocol/Port)

If you are unauthorized → Entry denied 🚫

🔧 3 Extended ACL Syntax

access-list <number> <permit|deny> <protocol> <source> <wildcard>
<destination> <wildcard> [eq <port_number>]

Key Notes:

- eq = equal to (port)
- gt = greater than
- lt = less than
- neq = not equal
- range = range of ports

Can use either **port number** or **well-known type** (e.g., HTTP, SSH)

📌 4 Where to Apply Extended ACL?

- Always **near the source** 🔑
 - Saves bandwidth by dropping unwanted traffic before it travels deep into the network
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5 Extended ACL Lab – Practical Examples

Network Setup:

- R1 connected to PCs
- R2 connected to Servers (SRV1, SRV2)
- IP addresses already configured

◆ Task 1 – Block 172.16.2.0/24 from communicating with PC1

```
R1(config)#access-list 110 deny ip 172.16.2.0 0.0.0.255 host 192.168.1.1
R1(config)#access-list 110 permit ip any any
R1(config)#interface g0/1
R1(config-if)#ip access-group 110 in
```

✓ Traffic from 172.16.2.0 network to PC1 is blocked

◆ Task 2 – Block 172.16.1.0/24 from accessing DNS on SRV1

```
R1(config)#access-list 110 deny tcp 172.16.1.0 0.0.0.255 192.168.1.100
0.0.0.255 eq 53
R1(config)#access-list 110 deny udp 172.16.1.0 0.0.0.255 192.168.1.100
0.0.0.255 eq 53
R1(config)#access-list 110 permit ip any any
R1(config)#interface g0/0
R1(config-if)#ip access-group 110 in
```

💡 Explanation:

Both TCP and UDP DNS traffic is blocked from the 172.16.1.0/24 network to SRV1 (192.168.1.100)

◆ Task 3 – Block 172.16.2.0/24 from accessing HTTP & HTTPS on SRV2

```
R1(config)#access-list 110 deny tcp 172.16.2.0 0.0.0.255 192.168.1.100
0.0.0.255 eq 80
R1(config)#access-list 110 deny tcp 172.16.2.0 0.0.0.255 192.168.1.100
0.0.0.255 eq 443
```

```
R1(config)#access-list 110 permit ip any any
R1(config)#interface g0/1
R1(config-if)#ip access-group 110 in
R1(config-if)#do wr
```

💡 Result:

- HTTP (80) and HTTPS (443) traffic is blocked from 172.16.2.0/24 network to SRV2
- All other traffic is allowed

Quick Notes on Wildcard Masks

Subnet Mask	Wildcard Mask
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255.255.255.0	0.0.0.255
---------------	-----------

255.0.0.0	0.255.255.255
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Wildcard mask = Reverse of subnet mask

Verification Commands

1 Check ACL on Interface

```
show ip interface g0/1
```

Look for:

Outgoing access list is 110 

2 Check ACL Hits

```
show access-lists
```

Example:

Extended IP access list 110

```
10 deny tcp 172.16.2.0/24 to 192.168.1.1 eq 80 (3 matches)
```

```
20 deny tcp 172.16.2.0/24 to 192.168.1.1 eq 443 (2 matches)
```

```
30 permit ip any any
```

Matches = Number of packets filtered or allowed

Key Points to Remember

- ✓ Extended ACL = Precise traffic control
 - ✓ Filters on **source, destination, protocol, port**
 - ✓ Placement = **near source** to save bandwidth
 - ✓ Default deny any → Always add permit ip any any at the end
 - ✓ Use eq, gt, lt, neq, range for ports
 - ✓ Verify using show access-lists and show ip interface
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Daily-Life Analogy

Extended ACL = Security guard with **advanced checklist**:

- Who you are (IP)
- Where you go (Destination IP)
- What you are carrying (Protocol/Port)

Traffic failing any rule → denied 

Traffic matching rules → allowed 

CHAPTER SUMMARY

1 Quick Port Revision (Very Important for Extended ACL)

Extended ACL works closely with **port numbers**, so remembering common ports is critical.

Port Protocol Service

20/21 TCP FTP

22 TCP SSH 

23 TCP Telnet

25 TCP SMTP

53 TCP/UDP DNS

67/68 UDP DHCP

69 UDP TFTP

80 TCP HTTP 

110 TCP POP3

123 UDP NTP

143 TCP IMAP

161 UDP SNMP

443 TCP HTTPS 

3389 TCP RDP

3306 TCP MySQL

5900 TCP/UDP VNC

 Ports = Doors in a building 

Extended ACL decides which door can be used.

2 What is Extended ACL?

An **Extended ACL** is a rule set that filters traffic based on:

- ✓ Source IP
- ✓ Destination IP
- ✓ Protocol (TCP, UDP, ICMP, IP)
- ✓ Port Number

It provides **precise and advanced filtering** compared to Standard ACL.

Extended ACL Features

Feature	Description
Filters on	Source + Destination + Protocol + Port
Placement	Near Source 
Number Range	100–199, 2000–2699
Flexibility	TCP, UDP, ICMP filtering
Control Level	High precision

Why Use Extended ACL?

- ✓ Block specific host to specific server
 - ✓ Allow only certain protocols
 - ✓ Block only HTTP but allow HTTPS
 - ✓ Restrict traffic by port number
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3 Extended ACL Syntax

```
access-list <number> <permit|deny> <protocol> <source> <wildcard>  
<destination> <wildcard> [eq port]
```

Port Operators:

- eq → Equal
- gt → Greater than
- lt → Less than
- neq → Not equal
- range → Port range

Placement Rule

 Extended ACL → Always place near the SOURCE

Why?

Because it drops unwanted traffic early and saves bandwidth.

Wildcard Mask Concept

Wildcard = Inverse of subnet mask

Subnet	Wildcard
255.255.255.0	0.0.0.255
255.0.0.0	0.255.255.255

Verification Commands

show access-lists

show ip interface g0/1

Hit count shows packet matches.

FINAL CONCLUSION

Extended ACL provides:

-  Advanced security
-  Port-level filtering

- ✓ Protocol-based control
- ✓ Efficient bandwidth usage

Standard ACL = Basic filtering

Extended ACL = Professional-level filtering 🧰

👉 Extended ACL is heavily used in enterprise networks.

Q & A

1 What is Extended ACL?

Extended ACL filters traffic based on source IP, destination IP, protocol, and port number.

It provides more precise control than Standard ACL.

2 Difference Between Standard and Extended ACL?

Standard	Extended
Source only	Source + Destination + Port
1–99	100–199
Near destination	Near source
Basic filtering	Advanced filtering

3 Why place Extended ACL near source?

To stop unwanted traffic early and save bandwidth.

4 What is implicit deny?

Every ACL ends with:

deny any

If no match → packet dropped.

5 What is wildcard mask?

Wildcard mask is inverse of subnet mask.

Used to match IP ranges in ACL.

6 What does eq mean?

eq = equal to specific port number.

Example:

eq 80

Blocks HTTP only.

7 How to block only HTTPS?

deny tcp any any eq 443

permit ip any any

8 How to block range of ports?

range 20 30

Blocks ports between 20 and 30.

9 How ACL processes rules?

Top to bottom.

First match wins.

Then stops processing

10 Why add permit ip any any at end?

To override implicit deny and allow remaining traffic.

1 1 Scenario: Block one subnet from accessing only web server but allow ping.

Solution:

```
deny tcp 172.16.2.0 0.0.0.255 host 192.168.1.10 eq 80
deny tcp 172.16.2.0 0.0.0.255 host 192.168.1.10 eq 443
permit icmp any any
permit ip any any
```

1 2 What happens if permit any any not added?

All unmatched traffic blocked due to implicit deny.

1 3 Can Extended ACL filter ICMP?

Yes. Example:

```
deny icmp any any
```

1 4 How to verify hit count?

```
show access-lists
```

Shows number of matches.

1 5 Can we use service name instead of port number?

Yes.

Example:

```
eq http
eq https
```

1 6 What happens if ACL applied wrong interface?

Traffic may bypass filtering or block wrong traffic.

1 7 Difference between inbound and outbound ACL?

Inbound → Filters before routing decision

Outbound → Filters after routing decision

1 8 Scenario: Block Telnet but allow SSH.

```
deny tcp any any eq 23
```

```
permit tcp any any eq 22
```

```
permit ip any any
```

1 9 How to remove Extended ACL?

```
no access-list 110
```

Remove from interface:

```
no ip access-group 110 in
```

2 0 What is named Extended ACL?

```
ip access-list extended BLOCK_WEB
```

More readable and editable.

PRACTICAL TROUBLESHOOTING QUESTIONS

2 1 Traffic not blocked even after ACL configured. Why?

Possible reasons:

✓ Wrong interface

✓ Wrong direction

- ✓ Wrong wildcard
 - ✓ Rule order incorrect
 - ✓ ACL not applied
-

2 2 ACL blocking too much traffic. Why?

Because implicit deny at end.

Need correct permit statements.

2 3 What is best practice for writing ACL?

- ✓ Specific rules first
 - ✓ General rules later
 - ✓ Add permit any any last
 - ✓ Comment ACL (if named)
-

2 4 What is difference between eq and range?

eq → single port

range → multiple ports

2 5 How Extended ACL helps in real enterprise?

- ✓ Protect servers
 - ✓ Block unauthorized apps
 - ✓ Restrict user internet usage
 - ✓ Secure internal network
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